

Symbolic methods of mathematical analysis

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Chapter 1.

Ring of power series

1.1 Power series

A natural generalization of the concept of a polynomial is the concept of a power series. Let t be a symbolic variable and k be some field, then expression of the form

$$\sum_{n=0}^{\infty} a_n t^n = a_0 + a_1 t + a_2 t^2 + \dots, \quad a_n \in k, \quad (1.1)$$

is called a power series or, longer, a series in powers of t . In this case, $a_0, a_1, \dots$ are called the coefficients of the series. For the series, different are considered different if they have at least one different coefficient.

Under the sum of two rows

$$\sum_{n=0}^{\infty} a_n t^n + \sum_{n=0}^{\infty} b_n t^n$$

understand a number of

$$\sum_{n=0}^{\infty} (a_n + b_n) t^n,$$

and under the product of series

$$\sum_{n=0}^{\infty} a_n t^n \cdot \sum_{n=0}^{\infty} b_n t^n$$

understand a number of

$$\sum_{n=0}^{\infty} c_n t^n,$$

the coefficients of which are calculated using the formula

$$c_n = a_n b_0 + a_{n-1} b_1 + \dots + a_0 b_n,$$

which is exactly the same as that used to calculate the coefficients of the product of two polynomials. For both addition and multiplication, to calculate the coefficient at a given power t , it is sufficient to perform a finite number of actions exactly.

Note: If desired, one can understand the power series as an infinite sequence of numbers from k and instead of the series (1.1) write $(a_0, a_1, a_2, \dots)$. Two such sequences are added element by element, and multiplied

according to a tricky rule:

$$(a_0, a_1, a_2, \dots) \cdot (b_0, b_1, b_2, \dots) = (c_0, c_1, c_2, \dots),$$

namely

$$c_n = \sum_{k=0}^n a_{n-k} b_k = \sum_{k=0}^n a_k b_{n-k}.$$

Example 1. Find the square of a series

$$1 + t + t^2 + t^3 + \dots$$

Note that all its coefficients are equal to 1 and therefore in the formulas for c_n one should be put in place of a_i and b_j . But then

$$c_n = \sum_{k=0}^n 1 = n + 1.$$

That's why

$$(1 + t + t^2 + t^3 + \dots)^2 = 1 + 2t + 3t^2 + \dots$$

or

$$\sum_{n=0}^{\infty} t^n = \sum_{n=0}^{\infty} (n+1)t^n$$

The set of all power series of t with coefficients from the field k with arithmetic operations introduced in this way is denoted as $k[[t]]$. This notation is also used in Sage, only Sage already knows that this set is ring.

```
sage : var ('t')
```

1

```
t
```

2

```
sage : QQ [[ t ]]
```

3

```
Power Series Ring in t over Rational Field
```

4

If we truncate all the terms of the series whose degree is greater than or is equal to n , then the series will yield a polynomial of degree t . We will call this truncation operation of a series and denote it using brackets. For example,

$$[3 + t^{25+t} + \dots]3 = 3 + t^2.$$

In Sage it is written like this

```
sage : QQ [[ t ]](3+ t ^2+ t ^5) . truncate (3)
```

5

```
t ^ 2 + 3
```

6

The definitions of the sum and product of power series are arranged so that calculation of the coefficient at t^n in total it is enough to know the coefficients at t^n the terms, therefore

$$[f + g]_n = [f]_n + [g]_n;$$

to calculate the coefficient at t^n in the product of series there are enough know the coefficients of $1, t, \dots, t^n$ in the factors, therefore

$$[f \cdot g]_n = [f \cdot [g]_n]_n = [[f]_n \cdot g]_n = [[f]_n \cdot [g]_n]_n.$$

These formulas allow us to reduce the proof of theorems about series to theorems on polynomials.

Theorem 1. The set of all power series of t with coefficients from fields k with arithmetic operations introduced in this way are a ring.

The proof of this theorem comes down to checking the axioms from the definition ring. Let us check for example the 4th axiom - associativity of multiplication-

zheniya

$$(f \cdot g) \cdot h = f \cdot (g \cdot h).$$

To do this, it is sufficient to prove that the coefficient of t^{n-1} on the left matches with a coefficient at t^{n-1} on the right. But the truncation of the left part is equal to

$$[(f \cdot g) \cdot h]_n = [(f \cdot g)_n \cdot h]_n = [(f \cdot g)_n \cdot h]_n = [(f \cdot g)_n \cdot h]_n,$$

and the truncation of the right side is equal to

$$[f \cdot (g \cdot h)]_n = [f \cdot (g \cdot h)_n]_n = [f \cdot (g \cdot h)_n]_n$$

These truncations are equal because the multiplication of polynomials is associative.

then the coefficients of $1, t, \dots$ the t^{n-1} on the left and right of the prov-equality coincide. Due to the arbitrariness of the choice of n , what has been said is true for all coefficients.

Traditionally, power series are written starting with the lowest-order term, that is, the term with a non-zero coefficient that has the smallest degree. The degree of this term in Sage is called the valuation. For in the left row this value is taken to be equal to $+\infty$:

```
sage : QQ [[ t ]](3* t ^6+ t ^2+ t ^5) . valuation() 7
2 8
sage : QQ [[ t ]](0) . valuation() 9
+ Infinity 10
```

The set of all power series whose valuation is greater than or equal to n are ideals of the ring $k[[t]]$ and are denoted by $O(t^n)$. For the proof

The fulfillment of two axioms from the definition of an ideal must be checked.

Two power series, say f and g , are equal in modulus $O(t^n)$ if

$$f - g \in O(t^n).$$

Traditionally instead

$$f = g \bmod O(t^n)$$

they write

$$f = g + O(t^n).$$

Looking at our definition of truncation, it is easy to see that

$$f = [f]_n + O(t^n).$$

Since there is no way to work directly with infinite sequences, in Sage, instead of a power series f , we consider the class $[f]_n + O(t^n)$, that is, an element of the quotient ring $k[[t]]/O(t^n)$. When we multiply two series

$$f = f_n + O(t^n), g = g_n + O(t^n)$$

as elements of the factor ring, we obtain

$$fng_n + O(t^n) = [fng_n]_n + O(t^n)$$

As we have already noted, the coefficients of the product fg at $1, t, \dots, t^{n-1}$ coincide with the coefficients at the same powers of $[fng_n]_n$. Now we could prove this by referring to the consistency of actions in the original ring and factor ring. However, we have already used this in proving proof that the set of power series is a ring.

A similar statement is also true for addition. Therefore, the transition from power series to elements of the factor ring $k[[t]]/O(t^n)$ allows one to perform arithmetic operations in a finite number of steps and gives a result that has the correct coefficients at powers of t that are less than n . Therefore, we say that we are working with series with an accuracy of up to terms of order n , and the quantity n itself is called precision.

If f is a polynomial in $k[t]$, then $k[[t]](f)$ is treated in Sage as a power series with a finite number of nonzero terms. Writing $k[[t]](f).O(n)$ turns it into an element of the quotient ring $k[[t]]/O(t^n)$. Thus, to find the square of the series from Example 1 in Sage, we need to define this series to some precision, that is, as an element of the quotient ring modulo $O(t^n)$,

and then square it:

```
sage : QQ [[ t ]]( sum ( t^ n for n in range (10) )) .O (5) ^2
```

11

```
1 + 2* t + 3* t ^2 + 4* t ^3 + 5* t ^4 + O (t ^5)
```

12

1.2. The inverse series to the given one

Let a series be given

$$f = f_0 + f_1 t + f_2 t^2 + \dots$$

Then the power series

$$x = x_0 + x_1 t + x_2 t^2 + \dots,$$

satisfying the equation

$$fx = 1,$$

called the inverse of f and write $x = f^{-1}$.

Theorem 2. If a power series is not equal to zero at $t = 0$, then there is an inverse power series.

Proof. We will look for a series inverse to a given one, not in the original ring of power series, but in the quotient rings $k[[t]]/O(t^n)$:

$$(f + O(t^n)) \cdot (x + O(t^n)) = 1 + O(t^n).$$

For $n = 1$ the statement is obvious:

$$f = f_0 + O(t), \quad x = \frac{1}{f_0} + O(t), \quad f \cdot x = 1 + O(t).$$

Let us assume that we have found a solution for some n , that is, such $x_0, \dots, x_{n-1}$, that

$$(f + O(t^n))(x_0 + x_1 t + \dots + x_{n-1} t^{n-1} + O(t^n)) = 1 + O(t^n).$$

We will look for a solution in factor rings $k[[t]]/O(t^{n+1})$ using the method of indefinite coefficient, taking

$$x = x_0 + x_1 t + \dots + x_{n-1} t^{n-1} + u t^n + O(t^{n+1}),$$

where u is an as yet undefined quantity. The coefficients of the product fx coincide at 1, $\dots$, with the coefficients of the same powers for the previous publications

$$(f + O(t^n))(x_0 + x_1 t + \dots + x_{n-1} t^{n-1} + O(t^n)) = 1 + O(t^n),$$

that is, the coefficient for 1 is equal to one, and the rest of the coefficients are zero. Coefficient-
coefficient at t^n in the work

$$(f + O(t^{n+1}))(x_0 + x_1 t + \dots + x_{n-1} t^{n-1} + u t^n + O(t^{n+1}))$$

is equal to the sum of the coefficient at the same degree in

$$(f + O(t^{n+1}))(x_0 + x_1 t + \dots + x_{n-1} t^{n-1} + O(t^{n+1}))$$

and the quantities $f_0 u$. By equating this coefficient to zero, we uniquely determine u and thereby find the solution to the equation $f x = 1$ in the factor ring $k[[t]]/O(t^{n+1})$. Starting with $n = 1$, we will

sequentially find the values of $x_1, x_2, \dots$

and as a result we get the following series

$$x = \frac{1}{f_0} + x_1 t + x_2 t^2 + \dots,$$

what is the coefficient of t^m the product $f x$ is equal to zero, that is

$$f x = 1.$$

This proves the theorem. □

The proof of the theorem was almost constructive: we can find
is clearly a coefficient at any given power t , but we cannot continue the process indefinitely.
Therefore, we can present x with an accuracy
to members of t^n , where n is some large number. For example,

sage : `QQ [[ t ]](1 - t) ^ -1` 13

`1 + t + t^2 + t^3 + t^4 + t^5 + t^6 + t^7 + t^8 + t` 14

`^9 + t^10 + t^11 + t^12 + t^13 + t^14 + t^15 + t`

`^16 + t^17 + t^18 + t^19 + O (t^20)`

sage : `QQ [[ t ]](1 - t + t^3) ^ -1` 15

`1 + t + t^2 - t^4 - 2* t^5 - 2* t^6 - t^7 + t^8 + 3* t` 16

`^9 + 4* t^10 + 3* t^11 - 4* t^13 - 7* t^14 - 7* t^15 -`

`3* t^16 + 4* t^17 + 11* t^18 + 14* t^19 + O ( t^20)`

In Sage, the default value is $n = 20$. In the documentation, this value is called accuracy (precision):

```
sage : QQ [[ t ]]. default_prec() 17
20 18
```

If desired, it can be redefined by defining a more expanded ring:

```
sage : PowerSeriesRing ( QQ , t , default_prec = 5) 19
Power Series Ring in t over Rational Field sage : PowerSeriesRing 20
(QQ, t , default_prec =5) (1 - t ) ^ -1 21
1 + t + t^2 + t^3 + t^4 + O(t^5) 22
```

In some cases, this infinite process can be described by a finite formula and thus write the answer in the original ring of power functions rows. For example,

$$f = 1 \div t,$$

Then

$$(1 \div t)(x_0 + x_1 t + x_2 t^2 + \dots) = 1$$

means that

$$x_0 + (x_1 \div x_0)t + (x_2 \div x_1)t^2 + \dots = 1$$

This gives

$$x_0 = 1, x_1 = x_0, x_2 = x_1, \dots$$

That's why

$$(1 \div t)^{-1} = 1 + t + t^2 + t^3 + \dots$$

As a consequence of the proven theorem, the ideal $O(t)$ is formed by all degrees series that do not have inverse elements in the ring $k[[t]]$.

Note: The ring $k[[t]]$ is an integral ring, so we can determine the field of fractions of this ring in which any one will be invertible non-zero element.

1.3. Differentiation of the ring of power series

In Algebra's course, differentiation D of the ring $k[t]$ was introduced:

$$D: \sum_{n=0}^N f_n t^n \mapsto \sum_{n=1}^N n f_n t^{n-1}.$$

Let us continue its continuation on $k[[t]]$, taking

$$D: \sum_{n=0}^{\infty} f_n t^n \mapsto \sum_{n=1}^{\infty} n f_n t^{n-1}.$$

This rule defines a linear mapping of the ring of power series onto itself. To call it a differentiation of the ring of power series, we must also prove that it satisfies Leibniz's rule.

$$D(fg) = gDf + fDg.$$

For this purpose, we compare the coefficients of t^n on the left side of this equality with the coefficient of the same power on the right-hand side. Differentiation lowers the powers by one, so this coefficient will not change if we truncate f, g to terms t^{n+2} . But for the polynomials $[f]_{n+2}$ and $[g]_{n+2}$, the Leibniz formula has already been proved in the algebra course, and therefore these coefficients coincide.

For a number of

$$f = f_0 + f_1 t + f_2 t^2 + \dots$$

the value is determined only at one point - at $t = 0$:

$$f(0) = f_0.$$

At any other point, the question of the meaning of the series leads us to the question of convergence of the series.

Theorem 3 (Taylor's). Let

$$f = f_0 + f_1 t + f_2 t^2 + \dots,$$

Then

$$f^{(n)} = n! f_n.$$

Proof: If we differentiate this series, we get

$$Df = f_1 + 2f_2t + 3f_3t^2 + \dots$$

From here we can find the derivative at zero: $Df(0) = f_1$. Differentiating once again, we get

$$D^2f = 2f_2 + 6f_3t + \dots,$$

from where $D^2f(0) = 2f_2$ etc. □

1.4. Equation $Dx = ax$ and exponential

An equation that connects the desired polynomial or series with its derivative called a differential equation. Most models in mechanics in science and physics, they are described by such differential equations. The simplest of differential equations is

$$Dx = ax, \quad a \neq 0.$$

Note: This equation with a negative real a describes all kinds of processes of cooling and decay, when the rate of decrease of x is proportional to x . For example, when describing the cooling of a liquid poured into glass, you can take x as the temperature of the liquid taken relative to ambient temperature, and assume that the rate of change of temperature is proportional to this temperature x . The hotter the liquid, the faster it cools.

The equation $Dx = ax$ cannot be solved in polynomials, since the degree of Dx is one less than the power of x . However, it is easily solved in power series. Substituting the expression

$$x = x_0 + x_1t + x_2t^2 + \dots$$

into the differential equation, we get

$$x_1 + 2x_2t + 3x_3t^2 + \dots = ax_0 + ax_1t + ax_2t^2 + \dots$$

From here we obtain recurrence relations for the coefficients:

$$x_1 = ax_0, x_2 = x_1, x_3 = x_2, \dots, x_n = \frac{a^n}{n!} x_0$$

From here it is easy to obtain explicit formulas:

$$x_n = \frac{a^n}{n!} x_0 = \frac{a^n}{n(n-1)\dots 1} x_0 = x_0 \frac{a^n}{n!}$$

Thus,

$$x = x_0 \left(1 + at + \frac{(at)^2}{2!} + \dots \right)$$

is a solution to the differential equation $Dx = ax$. Thus,

from, we got a solution in series containing one indefinite

constant - x_0 , which we will interpret as the value of x at

$t = 0$.

When we divide by n , we assume that in our field $n \neq 0$, which is not true in Galois fields.

From here on, we work in $\mathbb{Q}$ or its extensions $\mathbb{R}$, $\mathbb{C}$.

This simple thinking leads to several important for-
lectures.

Definition 1. Power series

$$\sum_{n=0}^{\infty} \frac{t^n}{n!} = \exp(t)$$

is called the exponential and is denoted as $\exp(t)$.

We can consider the expression $\exp(at)$ for any $a \in \mathbb{K}$ as a
formal series. In Sage, for example, $\exp(3t)$ can be defined as:

```
sage : QQ[[t]](3*t).exp()
1 + 3*t + 9/2*t^2 + 9/2*t^3 + 27/8*t^4 + 81/40*t^5 + 24
81/80*t^6 + 243/560*t^7 + 729/4480*t^8 +
243/4480*t^9 + 729/44800*t^10 + 2187/492800*t^11 +
2187/1971200*t^12 + 6561/25625600*t^13 +
```

23

$$\begin{aligned}
& 19683/358758400 \cdot t^{14} + 19683/1793792000 \cdot t^{15} + \\
& 59049/28700672000 \cdot t^{16} + 177147/487911424000 \cdot t^{17} \\
& + 59049/975822848000 \cdot t^{18} + 177147/18540634112000 \cdot \\
& t^{19} + O(t^{20})
\end{aligned}$$

Theorem 4. Differential equation

$$Dx = ax$$

has a solution equal to x_0 at $t = 0$:

$$x = x_0 \exp(at).$$

There are no other solutions in the ring $k[[t]]$ equal to x_0 at $t = 0$, these are differential

The equation does not have one.

Proof. Expression

$$x = x_0 \left(1 + at + \frac{(at)^2}{2!} + \dots \right)$$

is a solution because

$$Dx = x_0 D \left(1 + at + \frac{(at)^2}{2!} + \dots \right) = x_0 \left(a + + \dots \frac{at^2}{2!} \right) = ax.$$

If it had a different solution

$$x = x_0 + x_1 t + x_2 t^2 + \dots,$$

then, substituting it into $Dx = ax$, we would get

$$x_n = \frac{a}{n} x_{n-1} = \frac{a^2}{n(n-1)} x_{n-2} = \dots = \frac{a^n}{n(n-1)\dots 1} x_0 = \frac{a^n}{n!} x_0,$$

which means that this decision coincides with the one already indicated. □

Two very important consequences can be drawn from the proven theorem.

Theorem 5 (addition theorem). For any $a, b \in k$, the following holds:

$$\exp(at) \exp(bt) = \exp((a+b)t).$$

Proof. Let us assume

$$x = \exp(at) \exp(bt),$$

then by virtue of Leibniz's rule

$$Dx = D \exp(at) \cdot \exp(bt) + \exp(at) \cdot D \exp(bt).$$

By virtue of Theorem 4

$$D \exp(at) = a \exp(at), D \exp(bt) = b \exp(bt),$$

That's why

$$Dx = (a + b) \exp(at) \cdot \exp(bt) = (a + b)x.$$

Since $x(0) = 1$, due to the uniqueness of the solution of the equation $Dx = ax$, established in Theorem 4,

$$x = \exp((a + b)t),$$

that and so on. □

In Sage, for example, $\exp(3t)$ can be defined as:

```
sage : QQ [[ t ]]( t ) . exp () * QQ [[ t ]](2* t ) . exp()
1 + 3* t + 9/2* t ^2 + 9/2* t ^3 + 27/8* t ^4 + 81/40* t ^5 + 26
81/80* t ^6 + 243/560* t ^7 + 729/4480* t ^8 +
243/4480* t ^9 + 729/44800* t ^10 + 2187/492800* t ^11 +
2187/1971200* t ^12 + 6561/25625600* t ^13 +
19683/358758400* t ^14 + 19683/1793792000* t ^15 +
59049/28700672000* t ^16 + 177147/487911424000* t ^17
+ 59049/975822848000* t ^18 + 177147/18540634112000*
t ^19 + O( t ^20)
```

25

```
sage : QQ [[ t ]](3* t ) . exp()
1 + 3* t + 9/2* t ^2 + 9/2* t ^3 + 27/8* t ^4 + 81/40* t ^5 + 28
81/80* t ^6 + 243/560* t ^7 + 729/4480* t ^8 +
243/4480* t ^9 + 729/44800* t ^10 + 2187/492800* t ^11 +
```

27

$$\begin{aligned}
& 2187/1971200 \cdot t^{12} + 6561/25625600 \cdot t^{13} + \\
& 19683/358758400 \cdot t^{14} + 19683/1793792000 \cdot t^{15} + \\
& 59049/28700672000 \cdot t^{16} + 177147/487911424000 \cdot t^{17} \\
& + 59049/975822848000 \cdot t^{18} + 177147/18540634112000 \cdot \\
& t^{19} + O(t^{20})
\end{aligned}$$

Theorem 6. For any $a \in \mathbb{K}$, the following is true:

$$\exp(at)^{-1} = \exp(-at).$$

Proof. The statement follows from the addition theorem for $b = -a$:

$$\exp(at) \exp(-at) = \exp((a - a)t) = \exp(0) = 1.$$

□

1.5. The equation $D^2x = ax$ and hyperbolic functions

Let us now turn to the simplest equation relating the second derivative, that is, acceleration, and x , namely

$$D^2x = ax.$$

Substituting the expression into this equation

$$x = \sum_{n=0}^{\infty} x_n t^n$$

we will receive

$$\sum_{n=0}^{\infty} n(n-1)x_n t^{n-2} = a \sum_{n=0}^{\infty} x_n t^n$$

From here we obtain recurrence relations for the coefficients:

$$n(n-1)x_n = a^2 x_{n-2}$$

From here it is easy to obtain explicit formulas:

$$x_n = \frac{a^2}{n(n-1)} x_{n-2} = \frac{a^4}{n(n-1)(n-2)(n-3)} x_{n-4} = \dots$$

If n is even, then in $n/2$ steps we arrive at x_0 :

$$x_n = \frac{a^n}{n!} x_0.$$

If n is odd, then we arrive at x_1 :

$$x_n = \frac{a^{n-1}}{n!} x_1$$

Thus,

$$x = x_0 \left(1 + \frac{(at)^2}{2!} + \frac{(at)^4}{4!} + \dots + \frac{(at)^{2n}}{(2n)!} \right) + \frac{x_1}{a} \left(at + \frac{(at)^3}{3!} + \frac{(at)^5}{5!} + \dots \right)$$

is a solution to the differential equation $Dx = ax$. Thus,

from, we got a solution in series containing two constants - x_0 and

x_1 , which can easily be interpreted as the values of x and Dx at $t = 0$.

Definition 2. Power series

$$\frac{a^3 t^3}{3!} + \frac{a^5 t^5}{5!} + \dots$$

is called the hyperbolic sine and is denoted as $\sinh(t)$, and the power row

$$\frac{a^2 t^2}{2!} + \frac{a^4 t^4}{4!} + \dots$$

is called the hyperbolic cosine and is denoted as $\cosh(t)$.

In Sage, for example, $\sinh(t)$ can be defined as:

`sage : QQ [[ t ]]( t ) . sinh()`

```
t + 1/6* t ^3 + 1/120* t ^5 + 1/5040* t ^7 + 1/362880* t ^9 30
+ 1/39916800* t ^11 + 1/6227020800* t ^13 +
1/1307674368000* t ^15 + 1/355687428096000* t ^17 +
1/121645100408832000* t ^19 + O( t ^20)
```

Theorem 7. Differential equation

$$D^2 x = a^2 x,$$

equipped with initial conditions

$$x(0) = x_0, Dx(0) = x_1,$$

has a solution

$$x = x_0 \cosh(at) + \frac{x_1}{a} \sinh(at).$$

This differential equation has no other solutions satisfying the same initial conditions in the ring $K[[t]]$.

Looking at the rows for $\sinh t$ and $\cosh t$, it is hard not to notice the pieces of the row for exponents. Series

$$\cosh t = 1 + \frac{t^2}{2!} + \frac{t^4}{4!} + \dots$$

consists of even terms of the series for the exponent, therefore

$$t = \frac{\exp(t) + \exp(-t)}{2} \cosh t.$$

Row

$$\sinh t = \frac{\exp(t) - \exp(-t)}{2} = t + \frac{t^3}{3!} + \frac{t^5}{5!} + \dots$$

consists of odd terms of the series for the exponent, therefore

$$= \frac{\exp(t) - \exp(-t)}{2} \sinh t.$$

These formulas are called Euler's formulas. They show that to solve a second-order differential equation, it is sufficient to have an exponential components.

1.6. Equation $D^2x + a^2x = 0$ and trigonometric functions

Let us now turn to the equation

$$D^2x + a^2x = 0.$$

This is a very popular mathematical model called the linear oscillator.

Substituting the expression into this equation

$$x = \sum_{n=0}^{\infty} x_n t^n$$

we will receive

$$\sum_{n=0}^{\infty} n(n-1)x_n t^{n-2} = -a \sum_{n=0}^{\infty} x_n t^n$$

From here we obtain recurrence relations for the coefficients:

$$n(n-1)x_n = -a x_{n-2}$$

From here it is easy to obtain explicit formulas:

$$x_n = -a \frac{x_{n-2}}{n(n-1)} = +a \frac{x_{n-4}}{n(n-1)(n-2)(n-3)} = \dots$$

If n is even, then in $n/2$ steps we arrive at x_0 :

$$x_n = \frac{(-1)^{n/2} a^{n/2}}{n!} x_0.$$

If n is odd, then we arrive at x_1 :

$$x_n = \frac{(-1)^{(n-1)/2} a^{(n-1)/2}}{n!} x_1$$

Thus,

$$x = x_0 \left(1 - \frac{a^2 t^2}{2!} + \frac{a^4 t^4}{4!} - \dots \right) + x_1 \left(t - \frac{a^3 t^3}{3!} + \frac{a^5 t^5}{5!} - \dots \right)$$

is a solution to the differential equation $Dx = ax$. Thus,

from, we got a solution in series containing two constants - x_0 and

x_1 , which can easily be interpreted as the values of x and Dx at $t = 0$.

Definition 3. Power series

$$\sin(t) = t - \frac{t^3}{3!} + \frac{t^5}{5!} - \dots$$

is called sine and is denoted as $\sin(t)$, and the power series

$$\cos(t) = 1 - \frac{t^2}{2!} + \frac{t^4}{4!} - \dots$$

is called the cosine and is denoted as $\cos(t)$.

In Sage, for example, $\sin(t)$ can be defined as:

`sage : QQ [[ t ]]( t ) . sin()`

31

$$t - \frac{1}{6}t^3 + \frac{1}{120}t^5 - \frac{1}{5040}t^7 + \frac{1}{362880}t^9 - \frac{1}{39916800}t^{11} + \frac{1}{6227020800}t^{13} - \frac{1}{1307674368000}t^{15} + \frac{1}{355687428096000}t^{17} - \frac{1}{121645100408832000}t^{19} + O(t^{20})$$

Theorem 8. Differential equation

$$D^2x + ax = 0,$$

equipped with initial conditions

$$x(0) = x_0, Dx(0) = x_1,$$

has a solution

$$x = x_0 \cos(at) + \frac{x_1}{a} \sin(at).$$

This differential equation has no other solutions satisfying the same initial conditions in the ring $k[[t]]$.

If we work on $\mathbb{C}$, we can take $a = ib$ and reduce the differential equation to $D^2x = b^2x$, discussed earlier. In this case,

$$\cosh it = 2 \frac{\exp(it) + \exp(-it)}{2} \cos t =$$

And

$$1 \sin t = \sinh it = i \frac{\exp(it) - \exp(-it)}{2i}$$

These formulas are called Euler's formulas. They show that to solve this second-order differential equation it is sufficient to

but exhibitors, but for this you will have to pay by entering the complex plane.

1.7. Other differential equations

Differential equation

$$F(D^n x, D^{n-1} x, \dots, x, t) = 0$$

can be solved in power series. However, in practice the same problem arises a problem we have already encountered when inverting power series.

We can describe a procedure that, from a solution found with exactly t^n , yields a solution to member t^{n+1} , accurate to terms t^{n+1} . We can

apply this procedure many times and find a solution with some accuracy

ity, but we cannot transform the formulas for recalculating the new coefficient using the old ones into an explicit formula for calculating the coefficient.

This trick works only for a very narrow range of differential equations.

equations.

Sage has a built-in function to solve the equation

$$Df = a(t)f + b(t),$$

where a, b are polynomials or series in powers of t .

Example 2. Let's find a solution to the initial problem

$$Df = (2 - 3t + 4t^2)f + (3 - 4t^2), f(0) = 3.$$

Let's define the coefficients a, b of the equation and apply the method

`solve_linear_de`:

```
sage : a = QQ [[ t ]](2 - 3* t + 4* t ^2) sage : b = QQ [[ t ]] 33
```

```
(3 - 4* t ^2) sage : a . solve_linear_de ( b=b 34
```

```
, , prec = 15) f0 = 3 35
```

```
3 + 9* t + 9/2* t ^2 - 10/3* t ^3 + 95/24* t ^4 + 431/60* t 36
```

```
^5 - 1301/720* t ^6 - 3359/2520* t ^7 + 158701/40320* t
```

```
^8 + 18721/36288* t ^9 - 5845291/3628800* t ^10 +
```

```
258607/259200* t ^11 + 32265269/43545600* t ^12 -
```

```
173190679/283046400* t ^13 + 61616509/1585059840* t
```

```
^14 + O( t ^15)
```

Homework

Theoretical tasks

- 1) How are arithmetic operations defined in the ring of power series?
 $k[[t]]$?
- 2) Prove that the set of power series is a ring?
- 3) Prove that the set $O(t)$ of series $k[[t]]$ is an ideal of the ring of power numbers
- 4) What is the valuation of a power series?
- 5) What is the precision of a power series?
- 6) Under what conditions does a series have an inverse?
- 7) Give definitions of the series $\exp t$, $\sinh t$, $\sin t$.
- 8) Formulate the addition theorem for exponentials.
- 9) Using Euler's formula, prove:
 - a) the fundamental hyperbolic identity $\cosh^2 t - \sinh^2 t = 1$, b) the fundamental trigonometric identity $\sin^2 t + \cos^2 t = 1$,

Practical tasks

- 1) Let

$$f = \sum_{n=0}^{\infty} t^n, g = \sum_{n=0}^{\infty} n t^n$$

- a) Find the coefficient of t^5 in the expression $f + g$
- b) Find the coefficient of t^5 in the expression fg
- c) Find the coefficient of t^{50} in the expression fg
- d) Find the first 15 terms of the series for $1/f$.

2) Calculate the estimate of the power series

$$2t + 1 \ddot{y} = \frac{1}{2} - \cos t.$$

3) Truncate the series $\sin 3t$ to t^5 .

4) Find the first 10 terms of the series $\tan t = \sin t / \cos t$.

5) Find the solution to the initial problem over field A

a) $Dx + 2x = 0, x(0) = 0,$

b) $Dx + 2x = 0, x(0) = 3,$ c) $D^2x +$

$2x = 0, x(0) = 0, Dx(0) = 0,$ d) $D^2x + 2x = 0, x(0) = 1,$

$Dx(0) = 3,$ e) $D^2x - 2x = 0, x(0) = 1, Dx(0) = 3,$

6) Above field C, express the solutions of the previous problem through exponentials.

You.

Chapter 2.

Exponent

2.1 Numerical experiments with exponential

In the last chapter we found a solution to the differential equation

$$Dx = x$$

in the form of a formal power series

$$\exp t = 1 + t + \frac{t^2}{2!} + \frac{t^3}{3!} + \dots = \sum_{n=0}^{\infty} \frac{t^n}{n!} \quad (2.1)$$

If we substitute any value in place of t , say, $t = 1$, then we get

the sum of a series of rational numbers is calculated, namely

$$\exp 1 = 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \dots = \sum_{n=0}^{\infty} \frac{1}{n!}$$

Such an expression is called the sum of a numerical series or simply a numerical one.

We can't combine an infinite number of numbers, but we can substitute in place of infinity, various large integers:

<code>sage : sum ([1/ factorial (n) for n in range (10)])</code>	37
98641/36288	38
<code>sage : sum ([1/ factorial (n) for n in range (20)])</code>	39
82666416490601/30411275102208	40

To compare the resulting fractions, we convert them to decimals:

```
sage : RR ( sum ([1/ factorial ( n ) for n in range (10) ]) ) 41
2.71828152557319 42
sage : RR ( sum ([1/ factorial ( n ) for n in range (20) ]) ) 43
2.71828182845905 44
sage : RR ( sum ([1/ factorial ( n ) for n in range (30) ]) ) 45
2.71828182845905 46
sage : RR ( sum ([1/ factorial ( n ) for n in range (42) ]) ) 47
2.71828182845905 48
```

It is easy to see that with 20 terms or more the same number is obtained.

Of course, if you increase the number of bits allocated to the decimal fraction,

You can notice a slight difference in these amounts:

```
sage : RealField (70) ( sum ([1/ factorial ( n ) for n in 49
    range (20) ]) )
2.7182818284590452349 50
sage : RealField (70) ( sum ([1/ factorial ( n ) for n in 51
    range (30) ]) )
2.7182818284590452354 52
sage : RealField (70) ( sum ([1/ factorial ( n ) for n in 53
    range (42) ]) )
2.7182818284590452354 54
```

Thus, we cannot calculate the infinite sum exactly, but with

with a small margin of error we can write

$$\exp 1 \approx 2.718281828459045235360287471352 \dots$$

For other values of t we will see approximately the same thing. For example,
at $t = 2$

```
sage : RealField (70) ( sum ([2^ n/ factorial ( n ) for n in 55
    range (20) ]) )
```

7.3890560989301740963	56
<code>sage : RealField (70) (sum ([2^ n/ factorial (n) for n in range (30)]))</code>	57
7.3890560989306502272	58
<code>sage : RealField (70) (sum ([2^ n/ factorial (n) for n in range (42)]))</code>	59
7.3890560989306502272	60

Thus, we get approximately the same number:

$$\exp 2 \approx 7.38905609893065022723042313404 \dots$$

This allows us to consider the power series of the exponential as a function in the sense of school mathematics, namely, as a rule, which puts in the so-answer to the number t number

$$\sum_{n=0}^{\infty} \frac{t^n}{n!}.$$

2.2. Cauchy criterion

The basic idea underlying the theory of real numbers is the following: if something can be calculated with any given accuracy, then this something exists. An example of such a statement is the required We now need the Cauchy criterion.

Let there be a series of numbers

$$a_0, a_1, a_2, \dots$$

from $\mathbb{R}$ or $\mathbb{C}$, and let for any positive number $\varepsilon > 0$ there be such N , that the sums

$$\sum_{n=0}^M a_n$$

for $M \geq N$ differ from each other by no more than ε . In this case,

there is a number α such that the sums

$$\sum_{n=0}^M a_n$$

for $M \leq N$ they differ from this number by no more than ϵ .

In other words, if the amounts differ from each other by an arbitrary amount small, then you can indicate a number that differs from these sums of products.

freely a little.

For convenience, let us break down the Cauchy criterion into parts. If for any positive number $\epsilon > 0$ there is N such that the sums

$$\sum_{n=0}^M a_n$$

if $M \leq N$ differ from each other by no more than ϵ , then we say that row

$$a_0 + a_1 + a_2 + \dots$$

converges. If for any positive number $\epsilon > 0$ there is such an N , what amounts

$$\sum_{n=0}^M a_n$$

for $M \leq N$ differ from the number A by no more than ϵ , then we say that the series

$$a_0 + a_1 + a_2 + \dots$$

converges to a number A and we write

$$a_0 + a_1 + a_2 + \dots = A$$

The Cauchy criterion states that a convergent series necessarily converges to some number.

As a consequence of these definitions, we will indicate a very simple necessary convergence criterion.

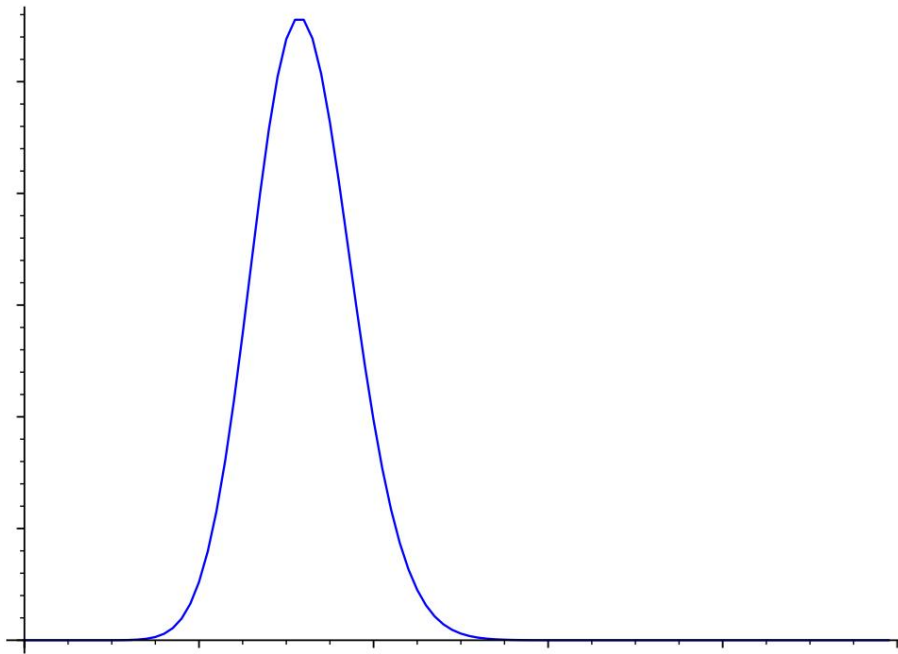


Fig. 2.1. Graph of the dependence of the terms of the series $\exp t$ at $t = 32$ on the term number, the scale along the ordinate is 1012

Theorem 9 (necessary convergence test, Euler's test). If the series

$$a_0 + a_1 + \dots$$

converges, then $|a_n|$ becomes arbitrarily small as n increases.

Proof. By the assumption, for any $\epsilon > 0$ there exists N such that

$$\sum_{n=0}^M a_n, \quad \sum_{n=0}^{M+1} a_n, \quad (M > N)$$

differ from each other by no more than ϵ . But the difference between these sums is equal to a_{M+1} . Therefore, for any ϵ there exists an N such that $|a_{M+1}| < \epsilon$ only $M > N$. □

Often, in practical calculations, the sum of several terms is calculated row and, noticing that the next member is too small or, on the contrary, too is great, they rush to draw a conclusion about the convergence or divergence of the series. Such kinds of judgments can be wrong.

Example 3. The terms of the exponential series at $t = 32$, when $n = 0$ changes to 30, grow quickly and quite radically, reaching values of the order of 10^{12} . However, this does not in any way hinder the convergence of the series.

2.3. Geometric progression series

Number series

$$1, q, q^2, q^3, \dots$$

is called a geometric progression. When n increases without limit, tends to infinity or to q^n zero, depending on whether $|q|$ is greater than or less than one. According to the Euler criterion, the series

$$1 + q + q^2 + \dots$$

diverges for $|q| \geq 1$. Let us

turn to the case $|q| < 1$. Since

$$(1 + q + \dots + q^{M-1})(1 - q) = 1 + q + \dots + q^{M-1} - q - q^2 - \dots - q^M = 1 - q^M,$$

right

$$1 + q + \dots + q^{M-1} = \frac{1 - q^M}{1 - q}.$$

From this follows the inequality

$$|1 + q + q^2 + \dots + q^{M-1}| < \frac{1}{1 - q}. \quad (2.2)$$

It is important for us that the right-hand side of this inequality does not depend on M .

Let N be a natural number and numbers M_1, M_2 are greater than N , and $M_2 > M_1$, then the difference

$$1 + q + \dots + q^{M_1-1}$$

And

$$1 + q + \dots + q^{M_2-1}$$

can be estimated by absolute value as

$$q^{M_1} + \dots + q^{M_2-1} = q^{M_1}(1 + q + \dots) < \frac{q^{M_2}}{1 - |q|} - \frac{q^{M_1}}{1 - |q|}.$$

Thus, this estimate is independent of M_1 and M_2 .

Since for $|q| < 1$ the value of $|q|^n$ tends to zero for any $\epsilon > 0$ it is possible to choose N such that

$$\frac{|q|^N}{1 - |q|} < \epsilon.$$

Therefore, by the Cauchy criterion, there exists such a real number A , what

$$1 + q + \dots + q^{M-1}$$

differs from A arbitrarily little for all sufficiently large M , then

There is

$$1 + q + q^2 + q^3 + \dots = A.$$

Of course, from

$$1 + q + \dots + q^{M-1} = \frac{1 - q^M}{1 - q}$$

it's immediately obvious that

$$1 + q + q^2 + q^3 + \dots = \frac{1}{1 - q}.$$

In other words, the geometric progression series converges for $|q| < 1$ and its the sum is equal

$$1 + q + q^2 + q^3 + \dots = \frac{1}{1 - q}.$$

Example 4. Let us illustrate the convergence with calculations for $q = \frac{1}{3}$.

sage : q = 1/3	61
sage : RR (1/(1 - q))	62
1.5000000000000000	63
sage : RR (sum ([q^n for n in range (10)]))	64
1.49997459736829	65
sage : RR (sum ([q^n for n in range (20)]))	66

1.49999999956980

67

sage : RR (sum ([q^ n for n in range (30)]))

68

1.499999999999999

69

sage : RR (sum ([q^ n for n in range (42)]))

70

1.500000000000000

71

Note: In the last chapter we saw that in the ring $\mathbb{Q}[[t]]$ the inverse of $1 - t$ is $\sum_{n=0}^{\infty} t^n$. Now we see that this series $\sum_{n=0}^{\infty} \frac{t^n}{n!}$ converges to e^t .

converges to $\frac{1}{1-t}$.

2.4. The number e

As a basic example, let's consider a number series

$$\sum_{n=0}^{\infty} \frac{1}{n!} = 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \dots, \quad (2.3)$$

which is obtained from the power series $\exp t$ at $t = 1$. Let us note its properties state.

1. Series (2.3) converges. To prove this, we use the estimate

$$\frac{1}{n!} < \frac{1}{1 \cdot 2 \cdot 2 \dots 2} = \frac{1}{2^{n-1}}.$$

Let N be a natural number and numbers M_1, M_2 are greater than N , where $M_2 > M_1$,

then the difference

$$\sum_{n=0}^{M_1} \frac{1}{n!}$$

And

$$\sum_{n=0}^{M_2} \frac{1}{n!},$$

that is

$$\sum_{n=M_1+1}^{M_2} \frac{1}{n!},$$

can be estimated by absolute value as

$$\sum_{n=M_1+1}^{M_2} \frac{1}{2^{ny_1}} = \frac{1}{2^{M_1+1}} \frac{1}{2^{ny_1}} (1 + \frac{1}{2^{ny_1}} + \dots) < (1 + \frac{1}{2^{ny_1}}) \frac{1}{2^{M_1+1}}$$

Applying estimate (2.2) for $q = \frac{1}{2}$, we have

$$\frac{1}{2^{N+1}} \frac{1}{2^{ny_1}} = \frac{1}{2^{N+1}}$$

It is obvious that for any $\epsilon > 0$ there exists N such that

$$\frac{1}{2^{N+1}} < \epsilon.$$

With this choice of N and any M_1 and M_2 , the difference

$$\sum_{n=0}^{M_1} \frac{1}{n!}$$

And

$$\sum_{n=0}^{M_2} \frac{1}{n!}$$

will be less than ϵ . This means that the sum (2.3) converges to some real number. This number is called the number e in honor of Euler and are presented as the value of the power series $\exp t$ at $t = 1$.

2. The number e lies between 2 and 3:

$$2 < e < 3.$$

The first inequality follows immediately from the definition:

$$1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \dots > 2 + \frac{1}{2!} = \frac{5}{2} > 2.5$$

To prove the first inequality it is enough to estimate the terms of the series from above like

$$\frac{1}{n!} < \frac{1}{1 \cdot 2 \cdot 2 \cdot \dots \cdot 2} = \frac{1}{2^{n-1}},$$

Then

$$1 + e = 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \dots < 3 + \frac{1}{2!} + \frac{1}{3!} + \dots < 3 + \frac{1}{2} + \frac{1}{4} + \dots < 3 + \frac{1}{2} = \frac{7}{2} < 4,$$

the row on the left is geometric with $q = 1/2$, therefore

$$e < 1 + 1 + \frac{1}{2} = 3.$$

3. The number e can be estimated more accurately as follows:

$$\sum_{n=0}^N \frac{1}{n!} < e < \sum_{n=0}^N \frac{1}{n!} + \frac{2}{(N+1)!}.$$

To prove this, we note that the remainder of the series

$$\sum_{n=N+1}^{\infty} \frac{1}{n!} = \frac{1}{(N+1)!} \left(1 + \frac{1}{N+2} + \frac{1}{(N+2)(N+3)} + \dots \right)$$

can be estimated from above as

$$\frac{1}{(N+1)!} \left(1 + \frac{1}{N+2} + \frac{1}{(N+2)^2} + \dots \right) = \frac{1}{(N+1)!} \frac{1}{1 - \frac{1}{N+2}} = \frac{2}{(N+1)!}.$$

For example, to calculate e with an accuracy of 10^{-2} , it is enough to take $N = 5$ members:

```
sage : 2/ factorial (4+1).n () 72
0.01666666666666667 73
sage : 2/ factorial (5+1).n () 74
0.00277777777777778 75
```

In this case, the number e can be estimated as follows:

```
sage : sum ((1/ factorial ( m ) ) .n () for m in range (5+1) 76
    ])
2.716666666666667 77
sage : sum ((1/ factorial ( m ) ) .n () for m in range (5+1) 78
    ]) + 2/ factorial (5+1)
2.719444444444444 79
```

In particular, the weaker statement follows from this:

$$e = 2.71 \pm 0.01.$$

More correct digits can be calculated in the same way. The constant e is usually
 Defense in CAS:

```
sage : e .n () 80
2.71828182845905 81

sage : sum ([ (1/ factorial ( m) ) .n () for m in range (17) ]) 82
2.71828182845904 83

sage : sum ([ (1/ factorial ( m) ) .n () for m in range (18) ]) 84
2.71828182845905 85
```

4. The number e is irrational. Let us assume the opposite: let e be a rational number. Then there exists an integer $N > 1$ so large that $N!e$ is an integer. But by the previous

$$\sum_{n=0}^N \frac{N!}{n!} < N!e < \sum_{n=0}^N \frac{N!}{n!} + \frac{2}{N+1}.$$

All terms on the left are numbers $\frac{N!}{n!}$ are integers, so there are two numbers n!

$$N!e \text{ and } \sum_{n=0}^N \frac{N!}{n!}$$

the difference between which is less

$$\frac{2}{N+1} < 1,$$

which is impossible.

Note: The above proof of the irrationality of the number e is due to Fourier (1815).¹

Attempts have been made for a long time

express the number e through integers using radicals or some

more algebraic operations. But it turned out that it is impossible to find an algebraic

equation with integer coefficients whose root would be

number e. The proof of this circumstance was discovered comparatively

¹On squaring the circle. With an appendix on the history of the question, compiled by F. Rudio. M.-L., 1936. P. 78.

recently in 1874 by Charles Hermite (1822–1901); the classic proof can be found in Klein², modernized - by S.

Leng³. A number that is not a root of any algebraic number.

equations with integer coefficients are called transcendental. The number

e became the first constant whose transcendence was rigorously proven.

Zana.

The elements of the series for the number e are rational numbers, the series is convenient satisfies the Cauchy criterion, but its sum is not a rational number.

crowbar. This means that the Cauchy criterion is not valid for rational

numbers. It follows that the proof of the Cauchy criterion for R relies on properties of the field R that are absent from the field Q. At the end of the 19th

various theories of real numbers were created in each of the centuries

which is based on a certain axiom from which the Cauchy criterion follows.

We will take it as an axiom. We will only note that from an algebraic point of view

From a purely theoretical point of view, the assertion of the existence of the number e is much more controversial than the assertion of the existence of the number i. The former relies on the proof of the completion of the field Q, for which the Cauchy criterion is valid, which goes beyond algebra, the second - on a purely algebraic proof-proof of the existence of a factor ring.

2.5. Power series as functions

Definition 4. Let the power series

$$S = \sum_{n=0}^{\infty} a_n t^n$$

converges for some numerical values of t that form a certain set G. We will say that this series defines a function whose domain of definition is the set G: this function puts into correspondence

²Klein F. Elementary Mathematics from the Higher Point of View. Vol. 1.

³Leng S. Algebra. Moscow: Mir, 1967. Addendum.

at a point $b \in G$ the sum of a numerical series

$$\sum_{n=0}^{\infty} a_n b^n,$$

which we will further denote as $s(b)$ or $s|_b$.

Let us show that the series

$$\exp t = \sum_{n=0}^{\infty} \frac{t^n}{n!}$$

we can consider as a function $\exp$ defined for all $t \in \mathbb{R}$ or even all $t \in \mathbb{C}$. By virtue of the Cauchy criterion, for this it is sufficient to prove, that for any given value of t and any positive value of ϵ there exists an N such that the sums

$$\sum_{n=0}^M \frac{t^n}{n!} \quad (2.4)$$

for $M \geq N$ they differ from each other by no more than ϵ .

Now let t be an arbitrary real or even complex number. Then there exists an integer K such that

$$K \leq |t| < K + 1,$$

let's put it this way

$$q = \frac{|t|}{K+1}, \quad 0 < q < 1.$$

For $n > K$

$$\frac{|t|^n}{n!} = \frac{|t|^K |t|^{n-K}}{1 \cdot 2 \cdot \dots \cdot (K+1) \cdot \dots \cdot n} < \frac{|t|^K}{n-K} q^{n-K}$$

Let $M_2 > M_1 \geq N > K$, then the sums

$$\sum_{n=0}^{M_1} \frac{t^n}{n!}$$

And

$$\sum_{n=0}^{M_2} \frac{t^n}{n!}$$

differ by no more than

$$\sum_{n=M_1+1}^{M_2} \frac{t^n}{n!},$$

which we can estimate from above as

$$\frac{|t|^K}{K!} + \dots + \frac{|t|^{M_2}}{M_2!} \leq \frac{|t|^{M_2}}{K!} \left(1 + \dots + |t|^{M_2-K} \right)$$

or, even more crudely, as

$$\frac{|t|^K}{K!} \frac{|t|^{M_2-K+1}}{1 - |t|}.$$

Since with the growth of n the expression $\frac{|t|^n}{n!}$ becomes less than any given q of the number, it is always possible to choose such a large $N > K$ that

$$\frac{|t|^K}{K!} \frac{|t|^{N-K+1}}{1 - |t|} < q$$

was less than the specified accuracy ϵ . But then for any $M_1, M_2 > N$ difference the number of the sums under consideration does not exceed ϵ .

Since in the series for hyperbolic and trigonometric functions contain the same members, but perhaps with different signs, written out the above estimates are also valid for them. Therefore, we can consider the series $\exp t$, $\sinh t$, $\cosh t$, $\sin t$, and $\cos t$ as functions defined for all $t \in \mathbb{C}$.

2.6. Arithmetic operations with power series

Let there be two power series, say,

$$f = \sum_{n=0}^{\infty} b_n t^n, \quad g = \sum_{n=0}^{\infty} c_n t^n$$

and let them converge at $t = a$. As elements of the ring of formal power series, they can be added and multiplied. The values of $f(a)$ and $g(a)$ can also be added and multiplied as elements of the field $\mathbb{R}$ or $\mathbb{C}$ under consideration. It seems obvious that the substitution operation $t = a$ and the arithmetic actions can be swapped:

$$(f + g)|_{t=a} = f|_{t=a} + g|_{t=a}, \quad (f \cdot g)|_{t=a} = f|_{t=a} \cdot g|_{t=a}.$$

However, in reality the second of these equalities can only be proven under additional assumptions about the nature of convergence.

The problem is this: For any $\varepsilon > 0$, of course, there exists such that for every $M \leq N$ and $t = a$

$$|[f]_M - f| < \varepsilon, |[g]_M - g| < \varepsilon,$$

where the square brackets mean the truncation of the series for $t = M$. From here it is easy to get an estimate

$$|fg - [f]_M[g]_M| = |fg - [f]_Mg + [f]_Mg - [f]_M[g]_M - [g]_M| + (|f| + \varepsilon)\varepsilon.$$

It is not difficult to choose ε for a given $\varepsilon > 0$ so that this value becomes less than ε . However, we are interested in

$$|fg - [fg]_M|,$$

that is, we need to estimate the difference

$$[fg]_M - [f]_M[g]_M,$$

which consists of terms of order $M+1, \dots, 2M$. The smallness of these terms does not follow from the convergence of the series f and g .

Definition 5. It is said that a number series

$$\sum_{n=0}^{\infty} a_n$$

converges absolutely if the series converges

$$\sum_{n=0}^{\infty} |a_n|.$$

Of course, if a series with modules satisfies the Cauchy criterion, then it satisfies a series without moduli. Therefore, from absolute convergence it follows. The usual blows, but not the other way around.

Theorem 10. Let two power series

$$f = \sum_{n=0}^{\infty} b_n t^n, g = \sum_{n=0}^{\infty} c_n t^n$$

converge absolutely at $t = a$. Then $f + g$ and $f \cdot g$ also converge at $t = a$.
absolutely and

$$(f + g)|_{t=a} = f|_{t=a} + g|_{t=a}, (f \cdot g)|_{t=a} = f|_{t=a} \cdot g|_{t=a},$$

The proof of this theorem is somewhat long and is given, for example, in the lecture by A. Hurwitz⁴.

To our great joy, in the proof of the convergence of the series for $\exp t$, $\sinh t$, $\cosh t$, $\sin t$ and $\cos t$, each term of the series is evaluated in absolute value, so in fact the absolute convergence of these series was proved.

2.7. Properties of the exponent

For the exponential as a power series, the addition theorem was proved

$$\exp(at) \exp(bt) = \exp((a + b)t).$$

The series $\exp(at)$ and $\exp(bt)$ converge absolutely at $t = 1$. By Theorem 10, the product $\exp(at) \exp(bt)$ converges absolutely at $t = 1$ to the value $\exp((a + b)t)$ at $t = 1$:

$$\exp(a) \exp(b) = \exp(a + b).$$

This is the addition theorem for the exponential in the usual school sense.

From it we can derive all other properties of the exponent.

1. For every $a \in \mathbb{C}$, the following is true:

$$\exp(a) \exp(-a) = 1.$$

2. For all real values of a , the following is true:

$$\exp(a) > 0.$$

⁴Gurvits A. Theory of functions. Ch. 1

For a ≥ 0 , all terms of the series $\exp a$ are positive and therefore their sum is also positive.

For a < 0 , by virtue of n.1

$$\exp(a) = \frac{1}{\exp(-a)} > 0.$$

particular,

the equation $\exp(t) = 0$ has no roots. 3. For $r \in \mathbb{Q}$, the following is true:

$$\exp(r) = e^r. \quad (2.5)$$

For natural values of the variable t , this formula immediately follows n. 1:

$$\exp(2) = \exp(1 + 1) = e \cdot e = e^2, \dots$$

If $t = m/n$, where n, m are natural numbers, then the positive number $\exp(m/n)$ satisfies the equation

$$x^n = e^m$$

by virtue of No. 1:

$$\exp(m/n)^n = \exp(m/n + \dots + m/n) = \exp(m) = e^m.$$

The positive root of this equation in elementary mathematics is m/n . Finally, if $t = \gamma m/n$, then by mean as e virtue of n. 1

$$\exp(\gamma m/n) = \exp(m/\frac{n}{\gamma}) = \frac{1}{\exp(m/n)^{\gamma}} = \frac{1}{e^{\gamma m/n}}.$$

Thus, formula (2.5) is true for all rational values of x . It is traditionally believed that formula (2.5) is valid for all those values of r for which series (2.1) converges, including all real values. In this case, the equality

$$e^x = \exp(x)$$

are taken as a definition.

2 Note: In the future, not only e will appear e^{-A} , but also e^A , where A is a square matrix. Of course, no one multiplies the number e A once.

2.8. Rounding error

If it is necessary to sum a large number of terms, a problem arises

It's a pleasure to turn each term into a decimal fraction and add them together.

already decimal fractions. This saves computer resources and time, but rounding errors accumulate.

For example, we can calculate $\sin 32$ as the sum of 100 terms of the series for sine:

```
sage : RR ( sum ([ (-1) ^n *32^(2* n +1) / factorial (2* n +1) for 86
              n in range (50) ]) )
0.551426671118916
```

87

This answer is correct:

```
sage : RR ( sin (32) )
0.551426681241691
```

88

89

However, if we convert the terms into decimal fractions immediately, we get what

```
sage : sum ([ RR (( -1) ^n *32^(2* n +1) / factorial (2* n +1) ) for n in range
              (50) ])
0.552008761698294
```

90

91

The error in the 3rd decimal place is clearly visible. This error is not related to truncation of the series by a hundred terms:

```
sage : sum ([ RR (( -1) ^n *32^(2* n +1) / factorial (2* n +1) ) for n in range
              (100) ])
0.552008771821069
sage : sum ([ RR (( -1) ^n *32^(2* n +1) / factorial (2* n +1) ) for n in range
              (200) ])
0.552008771821069
```

92

93

94

95

This is a rounding error. In Fig. 2.1 it is clear that among the terms of the series there are

very large numbers, which also come with different signs.

Homework

Theoretical tasks

- 1) Give the definition of convergence of a number series. What is the sum of the numbers?
left row?
- 2) Formulate the Cauchy criterion
- 3) Prove that $\sin 1$ is a real number.
- 4) Prove that $\sinh t$ converges for all real t .
- 5) Prove the basic trigonometric identity $\sin^2 a + \cos^2 a = 1$
for any real a
- 6) Prove that $|\sin a| \leq 1$ for any real a .
- 7) Prove the fundamental hyperbolic identity $\cosh^2 a - \sinh^2 a = 1$
for any real a
- 8) Prove that $e^{i} = \cos 1 + i \sin 1$.
- 9) Prove that the number of terms in the series for $\exp t$ required to calculate this expression for $|t| \leq 1$ with accuracy ϵ does not exceed the number of terms required to calculate e with the same accuracy.

Practical tasks

- 1) Calculate the number e using a series with an accuracy of 10^{-9} .
- 2) Calculate the number $\sin 1$ using a series with an accuracy of 10^{-5} .
- 3) Calculate $e^{2.1}$ using the addition theorem ($e^{2.1} = e^{2} \cdot e^{0.1}$).
- 4) Check the basic trigonometric identity for $t = 2$ by directly calculating the values of the series.

5) Check the main hyperbolic series values by direct calculation.

ical identity for $t = 2$.

6) How many terms should be taken in the series for $\cos 32$ to get that

the decimal fraction that $RR(\cos(32))$ gives?

7) (*) Sage has a built-in $\sin$ function. Does the design allow

$\text{RealField}(N)(\sin(1/10))$ calculates the $\sin$ value $\frac{1}{10}$ with an accuracy of

rounding error, which is determined by the number of bits allocated to

is this number?

Chapter 3.

Transcendental equations

3.1. Roots of the equation

Let f be some symbolic expression that depends on one variable x and for $a \leq x \leq b$ it takes real values. We assume that the graph of the function $y = f(x)$ on this segment can be drawn with one movement of the hand, without lifting the pencil from the paper. Such a function called continuous.

From this point of view it seems obvious that the following is true theorem.

Theorem 11. If a function f is continuous on a segment $a \leq x \leq b$, and $f(a)$ and $f(b)$ have different signs, then on this segment the equation $f(x) = 0$ has at least one real root.

For example, the equation

$$2^x = 2$$

has one root on the interval $1 < x < 2$, since $f(x) = 2^x - 2$ takes different signs at $x = 1$ and $x = 2$. This root is, of course, denoted as $\log_2 2$. A long time ago, a method was proposed that

allows one to find approximately the roots of the equation, the existence of which is guaranteed by Theorem 11.

3.2 Dichotomy

The easiest way to find the root of an equation

$$f(x) = 0$$

is called the method of dividing a segment in half and consists of the following.

we lay down

$$x_0 = a, y_0 = b$$

and divide this segment in half. If in the middle of the segment

$$x = \frac{x_0 + y_0}{2}$$

and at the left end $x = x_0$ the function has different signs, then the equation has the root on a segment whose ends are

$$x_0, y_1 = \frac{x_0 + y_0}{2}$$

otherwise the root is on the segment with ends

$$x_1 = \frac{x_0 + y_0}{2}, y_1$$

Dividing this segment in half again, we get the segment $x_2 < x < y_2$

lengths $\frac{b-a}{2^2}$, on which the root lies, etc. As a result, for one of the roots

the equation yields the following estimate:

$$x_n - y_n \leq \frac{b-a}{2^n}$$

Thus, in n moves we can find the root of the equation with an accuracy of $\frac{b-a}{2^n}$. By increasing n , we can to $\frac{b-a}{2^n}$ calculate the root with any given accuracy.

ness.

Listing 3.1. Calculating the root by dividing a segment in half

def dichotomy (f , a , b , epsilon): iflen

(f . variables ()) == 1: x = f . variables () [0]

```

else:
    return 'error'

iff . sub s ( [ x==a ] ) == 0:
    return a

iff . sub s ( [ x==b ] ) == 0:
    return b

iff. sub s ( [ x==a ] ) ÿ f . sub s ( [ x==b ] ) > 0:
    return 'error'

while abs ( b - a ) > 2 ÿ epsilon :
    c = (a + b) / 2
    iff . sub s ( [ x==c ] ) == 0:
        return c

    iff. sub s ( [ x==a ] ) ÿ f . sub s ( [ x==c ] ) < 0:
        b = c
    else:
        a = c

return (a + b) / 2

```

The implementation of the dichotomy method in Sage is presented in Listing 3.1. The arguments of the dichotomy function are f , the endpoints of the segment under consideration, a and b and the accuracy ϵ with which the root of the equation $f(x) = 0$ must be calculated. We, of course, do not store the elements of the series $x_0, x_1, \dots$ and $y_0, y_1, \dots$, and write down we write x_n as a , and y_n as b . The calculation of the elements of these series goes on until until the difference $y_n - x_n$ becomes less than 2ϵ . As soon as this difference became less than 2ϵ , the function returns the number $\frac{x_n + y_n}{2}$.

This number differs from the true root of the equation by no more than ϵ . in fact, by virtue of Theorem 11, on the segment $[x_n, y_n]$, the length of which is less than 2ϵ , there is a true root of the equation, which differs from the middle of this segment by no more than half its length, that is, less than ϵ .

Example 5. For equation $x^2 = 2$ on the segment $1 < x < 2$ with an accuracy of $\epsilon = 0.001$ we have:

```

sage : load ('sage / dichotomy . sage ') 96
None 97
sage : dichotomy ( x ^2 -2 ,1 ,2 ,0.001) 1449/1024 98
99
sage : RR ( dichotomy (x ^2 -2 ,1 ,2 ,0.001) ) 100
1.41503906250000 101
sage : RR ( sqrt (2) ) 102
1.41421356237310 103

```

It is clearly visible that we get the value $\sqrt{2}$ with an error in the 3rd digit after comma.

The dichotomy method is very unpretentious and allows, in particular, to obtain approximately the roots of algebraic equations of any degree. We can compare it with the standard one that we have used many times in the course algebra.

Example 6. For equation $x^5 + x + 1 = 0$ on the segment $\sqrt{2} < x < 2$ accurate to $\epsilon = 0.0001$ we have:

```

sage : f = x ^5 + x +1 104
sage : RR ( dichotomy (f , -2 ,2 ,0.0001) ) 105
-0.754821777343750 106
sage : RR ( RR ( dichotomy (f , -2 ,2 ,0.0001) ) ) 107
-0.754821777343750 108
sage : QQ [ x ]( f ) . roots (AA, multiplicities = False ) 109
[ -0.7548776662466928?] 110

```

It is clearly visible that we get a value for the root with an error in the 5th digit. after the decimal point, the 4th was determined correctly.

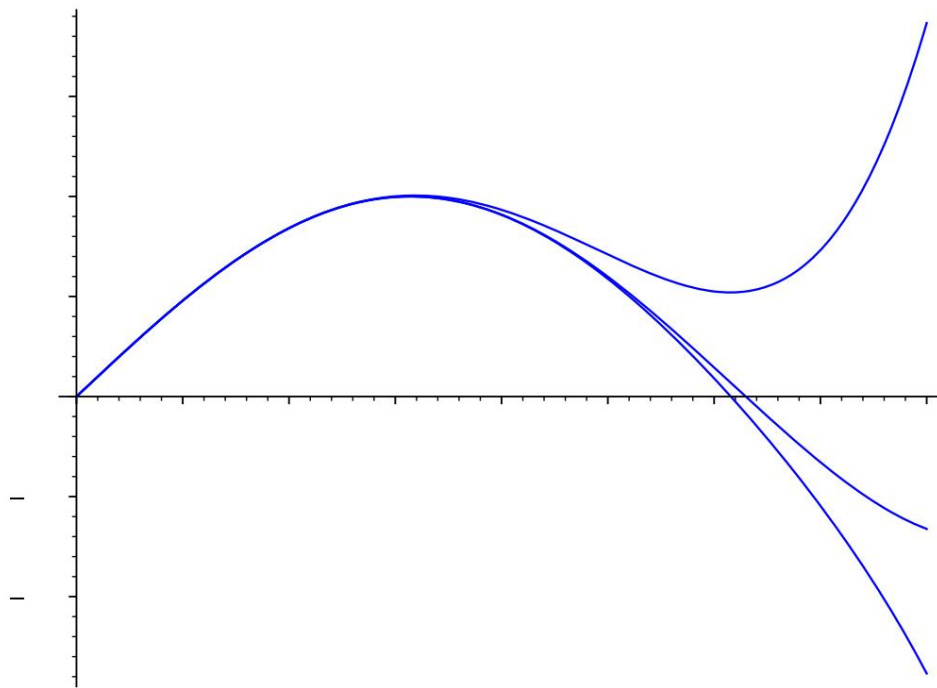


Fig. 3.1. Graph of the sums of the first 3, 4 and 5 terms of the series for $\sin x$.

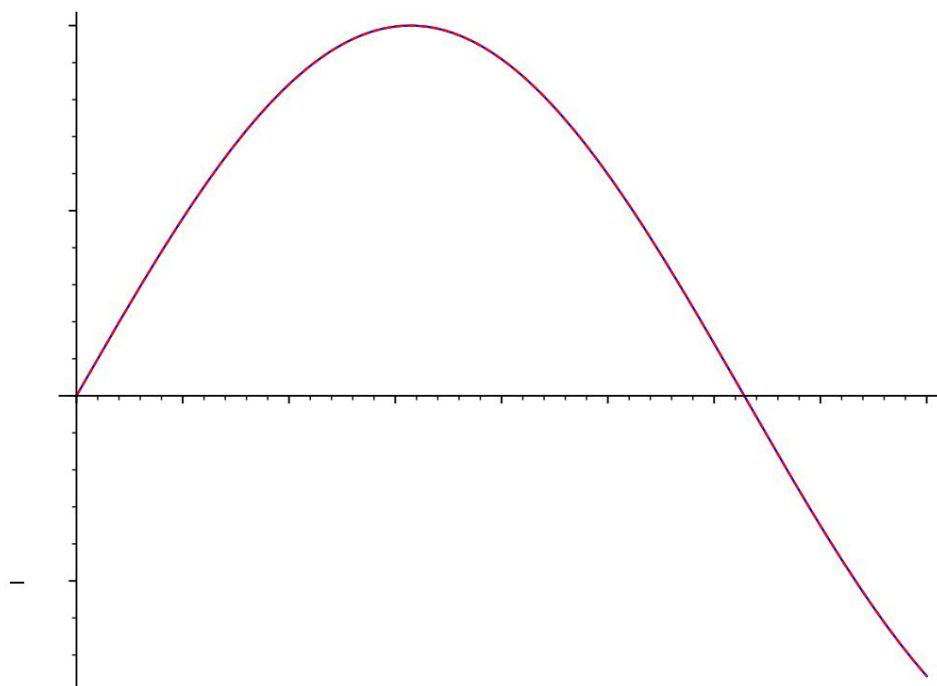


Fig. 3.2. Graph of $\sin x$ as a sum of 7 terms.

3.3 The number pi

Let us take by definition that the number π is the smallest positive root of the equation

$$\sin x = 0.$$

To calculate this number, we plot the sine graph on the interval $0 < x < \pi$.

For this purpose, let us recall that

$$\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!},$$

taking 3, 4 and 5 members in this series, we will construct graphs of the resulting multi-members, see Fig. 3.1. The characteristic scale is already clear from this figure.

On the ordinate axis, all values lie between 1 and a small negative number. Therefore, if we calculate the sine values with an accuracy of $\epsilon = 0.01$, we will not be able to notice this error on the graph.

Let $M_2 > M_1 \gg N$, then the difference of the sums

$$M_1 - M_2 = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$$

And

$$M_2 - M_1 = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$$

can be very roughly estimated in absolute value as

$$\sum_{n=N+1}^{\infty} \frac{4}{(2n+1)!}$$

and even rougher as

$$\sum_{n=2N+1}^{\infty} \frac{4^n}{n!} \approx e^4 - \sum_{n=0}^{2N} \frac{4^n}{n!}$$

Knowing the number e with good accuracy, we can choose N such that this value was less than the $\epsilon = 0.01$ given to us:

```
sage : RR ( e ^4 - sum ([4^ n/ factorial (n ) for n in range
(13) ]) ) 111
```

```
0.0149444321608598 112
```

```
sage : RR ( e ^4 - sum ([4^ n/ factorial (n ) for n in range
(14) ]) ) 113
```

```
0.00416739027270552 114
```

This means that 7 members will provide us with the required accuracy over the entire interval.

$0 < x < 4$:

$$\sin x \approx \sum_{n=0}^7 (-1)^n \frac{x^{2n+1}}{(2n+1)!}.$$

The graph of the polynomial on the right is shown in Fig. 3.2 and it, with graphical accuracy, coincides with the graph obtained with using Sage's built-in sin function. It's clear that, firstly, such a schedule can be drawn without lifting your hand from the paper, and, secondly, on the segment $3 < x < 4$ there is a root, which is called the number $\tilde{y}$.

Let's find it using the dichotomy method.

```
sage : f = sum ([(-1) ^ n* x ^(2* n +1) / factorial (2* n +1) for n
in range (7) ]) 115
```

```
sage : RR ( dichotomy (f ,3 ,4 ,0.1) ) 116
```

```
3.187500000000000 117
```

```
sage : RR ( dichotomy (f ,3 ,4 ,0.01) ) 118
```

```
3.148437500000000 119
```

```
sage : RR ( dichotomy (f ,3 ,4 ,0.001) ) 120
```

```
3.141601562500000 121
```

```
sage : RR ( dichotomy (f ,3 ,4 ,0.000001) ) 122
```

```
3.14161396026611 123
```

It is easy to see that we have obtained with some accuracy the known value calculations for the number $\tilde{y}$, however, increasing the accuracy of the dichotomy does not lead to clarification of the meaning. This happens because the expression used

for sine allows us to calculate sine with an accuracy that we estimated in $\delta = 0.01$. If the values of $|f(x_n)|$ or $|f(y_n)|$ become less than $\delta = 0.01$, then the sign of $f(x_n)$ or $f(y_n)$ may be determined with an error and we will choose a different interval. Therefore, in this case, the calculations must either be stopped, or refine the expression for sine.

However, this observation raises doubts about the correctness of applying Theorem 11 to the equation $\sin x = 0$. Where is the guarantee that when changing the scale of a continuous smooth sine graph does not turn into something very intricately discontinuous? What actually needs to be proven to apply the dichotomy method without a twinge of conscience?

3.4. Justification of the method of dividing a segment in half

Let's start with the fact that in the previous section we constructed two series of real numbers

$$x_0, x_1, x_2, \dots, y_0, y_1, y_2, \dots$$

They are often called sequences of numbers. We hope that both the sequences from the previous section converge to the root of the equation in the sense the following definition.

Definition 6. The set of points on the real line satisfying the inequality $|x - a| < \delta$ is called the δ -neighborhood of the point a .

Definition 7. A point a on the real line is called the limit of a sequence $\{a_n\}$ if outside any neighborhood of this point there are at least a finite number of points of the sequence. In this case,

jester

$$\lim_{n \rightarrow \infty} a_n = a \text{ or, in short, } a_n \rightarrow a$$

and imagine how, as n grows, the points a_n get closer and closer to a or, in other words, how the sequence converges to a .

For example, that the sequence

$$a_n = \frac{(-1)^n}{n}$$

has zero as its limit.

The definition does not say that the sequence can have a limit only one, since this is easily proved by contradiction. Suppose that the sequence $\{a_n\}$ has two distinct limits - a and a' of two points, one can specify disjoint neighborhoods U and U' . These have such that $a_n \rightarrow a$, outside U there are only a finite number of elements of the sequence. Since $a_n \rightarrow a'$, and, as soon as U and U' do not intersect, only these elements fall in U' . Therefore, point a' has a neighborhood outside of which lies infinity the number of elements in the sequence, which contradicts the definition limit.

The arithmetic properties of the limit operation can be collected into one theorem.

Theorem 12. If

$$a_n \rightarrow a \text{ and } b_n \rightarrow b,$$

That

$$a_n \pm b_n \rightarrow a \pm b$$

for any of the 4 arithmetic operations.

Proof. As an example, we will prove the statement of the theorem for multiplication. No matter how small the number $\epsilon > 0$ we take, all elements of the sequence a_n , starting from some number, fall into $|x - a| < \epsilon$, and all elements of the second sequence fall into $|x - b| < \epsilon$, therefore

$$|a_n b_n - ab| = |a_n(b_n - b) + (a_n - a)b| < (a + \epsilon)\epsilon + \epsilon(b + \epsilon) = (a + b + 2\epsilon)\epsilon.$$

For any neighborhood U of the point ab , one can choose such a small ϵ that region

$$|x - ab| < (a + b + 2\epsilon)\epsilon$$

fits into it entirely, and therefore, all the elements of the sequence lie in U
sequences a_n . □

Any numerical method for finding the root of the equation $f(x) = 0$ is a certain algorithm by which the elements of the sequence are found.

sequences $a_1, a_2, \dots$, and hope that when replacing the root a with a_N no noticeable error is made if N is large. The simplest way is the last property can be interpreted in the sense that

$$\lim_{n \rightarrow \infty} a_n = a$$

and $f(a) = 0$.

The fact that both sequences constructed using the dichotomy method are similar are being observed, it is easiest to see from the following principle.

Axiom 1 (on a monotone sequence). A monotone increasing sequence bounded above converges. Similarly, a monotone decreasing sequence bounded below converges.

The sequence $x_0, x_1, \dots$ is monotonically increasing:

$$x_{n+1} \geq x_n$$

and is bounded from above:

$$x_n \leq b.$$

Therefore, it has some limit c . Similarly, the sequence

$y_0, y_1, \dots$ monotonically decreases and is bounded from below by a constant a , therefore it has as its limit a certain number c .

Limits c and c both sequences coincide. Then they have

two disjoint neighborhoods. The first contains all points x_n , starting with a certain number, and the second contains all points y_n . But then the distance

between x_n and y_n there always remains greater than some positive value

ranks - the distance between the right border of the first neighborhood and the left

the second boundary. This contradicts the fact that $|y_n - x_n| = \frac{|b-a|}{2^n} \rightarrow 0$.

Thus we prove that both sequences have the same limit, we denote it as c .

It remains to prove that this limit is the root of the equation $f(x) = 0$. From the construction of the sequences it is obvious that

$$f(x_0), f(x_1), \dots$$

have the same sign, say plus. If the graph $y = f(x)$ is drawn with one hand movement, from $x_n \rightarrow c$ it follows that $f(x_n) \rightarrow f(c)$. Therefore, from $f(x_n) \rightarrow 0$ it follows that $f(c) = 0$. For the same reasons

$$f(y_0), f(y_1), \dots$$

have a different sign, so the limit of the sequence is $f(c) = 0$. This means that $f(c) = 0$.

Three important conclusions can be drawn from this.

First, the presented reasoning provides a proof of the theorem

11. In doing so, we have again transformed the ability to calculate something approximately as proof of its existence.

Secondly, we have proved the convergence of the dichotomy method under theoretical conditions.

Remy 11.

Thirdly, we saw how the concept of continuous-ness, without referring to the process of constructing a graph manually.

Definition 8. A function f is called continuous on a set M if for any sequence $\{a_n\}$ from $a_n \rightarrow a \in M$ it follows that $f(a_n) \rightarrow f(a)$.

Homework

Theoretical tasks

- 1) Formulate the definition of the limit of a sequence.
- 2) Formulate the definition of continuity of a function.

3) Formulate a theorem about the root of the equation $f(x) = 0$.

4) Describe the dichotomy method

Practical tasks

1) The equation is given

$$56x^5 - 75x^4 + 340x^3 - 675x^2 + 548x - 120 = 0$$

a) Find its roots in Sage using the root method.

b) Choose the interval $[a, b]$ so that you can find it using the dichotomy method smallest root.

c) Choose the interval $[a, b]$ so that you can find it using the dichotomy method the greatest root.

2) Equation

$$\prod_{n=0}^N (x - n) = 0$$

has only integer roots. However,

```
sage : f = SR ( ZZ [ x ] ( prod ([ x - n for n in range (30) ]) ) )
124 sage : RR ( dichotomy ( f
, 1.0 , 10.0 , 10^-10 ) )
125
7.95803225371492
126
```

Why is the error so much greater than the specified accuracy? How can this be corrected?
to twist?

3) Using the exponential series, find the root of the equation

$$e^x = 2$$

with an accuracy of $\epsilon = 0.01$. Compare with the exact solution $\ln 2$.

4) Using the cosine series, find the first positive root

equations

$$\cos x = 0$$

with an accuracy of $\epsilon = 0.01$. Compare with the exact solution.

- 5) With what maximum accuracy can the number π be calculated as a coefficient of the equation $\sin x = 0$ using the approximation

$$\sin x \approx \sum_{n=0}^{10} (-1)^n \frac{x^{2n+1}}{(2n+1)!}.$$

We believe that $\pi \cdot n(\text{digits}=10^2)$ gives the correct answer.

- 6) Is it possible to calculate π as the root of the equation $\sin x = 0$ with an accuracy of 10^{-10} using the built-in $\sin$ function in Sage?
- 7) Modify the dichotomy function so that the loop stops when the accuracy of the calculation of f is exhausted.
- 8) We know that π is the smallest positive root of the equation $\sin \pi = 0$. We also know the addition theorems for sine and cosine. Prove that:

a) $\cos \frac{\pi}{2} = 0$

b) $\sin \pi n = 0, n \in \mathbb{Z}$

c) $\sin(x + 2\pi) = \sin x$

d) $\cos(x + 2\pi) = \cos x$

- 9) (*) Prove that for $x \neq \pi n, n \in \mathbb{Z}$, the sine is not equal to zero.